

CS & IT ENGINEERING



Computer Network - 2

Transport & Application Layer

Lecture No. - 01

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Recap of Previous Lecture



Topic

NAT

Topic

Traffic Shaping



Topics to be Covered



Topic

DHCP

Topic

Socket Programming

ABOUT ME



Hello, I'm **Abhishek**

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[NAT] [GATE-2016, Set-1, 2-Marks]



#Q. For a host machine that uses the token bucket algorithm for congestion control, the token bucket has a capacity of 1 megabyte and the maximum output rate is 20 megabytes per second. Tokens arrive at a rate to sustain output at a rate of 10 megabytes per second. The token bucket is currently full and the machine needs to send 12 megabytes of data. The minimum time required to transmit the data is _____ seconds.

Ans = 1.1

Solution :-

Data Size = 12 MB

C = 1 MB (Megabytes) (Full)

M = 20 MBps

R = 10 MBps

S = Maximum duration for which
the computer can transmit at the full capacity

$$S = \frac{C}{(M-R)} = \frac{1 \text{ MB}}{(20-10) \text{ MB/sec}} = 0.1 \text{ sec}$$

$$\begin{aligned} \text{Minimum time required to transmit the data} &= S + \frac{[\text{Data Size} - (M \times S)]}{R} \\ &= 0.1 \text{ sec} + \frac{[12 \text{ MB} - 2 \text{ MB}]}{10 \text{ MB/sec}} = 1.1 \text{ sec} \end{aligned}$$

Suppose Token Bucket is initially empty



$$\text{Min}^m \text{ Time} = \frac{\text{Data Size}}{R} = \frac{12 \text{ MB}}{10 \text{ MB/sec}} = 1.2 \text{ sec}$$



Topic : IP Allocation



- When a host connect to network
- Unique IP address is assigned to it
[where NetID bits is already predefined]



Topic : IP Allocation



→ Two types of IP allocation policy :

1. Static Allocation

→ Manual update

2. Dynamic Allocation

→ Dynamic Host Configuration Protocol (DHCP)



Topic : DHCP



→ **DHCP** : Dynamic Host Configuration Protocol

→ Dynamic assignment of IP address
[IP address assignment on demand]

→ Avoid manual IP configuration

→ Support mobility of laptops

→ Earlier Protocols : RARP [Reverse ARP], BOOTP [Bootstrap Protocol]





Topic : DHCP



- Host dynamically obtains IP address from network server (DHCP server)
[when it join networks]
- Can renew its lease on address in use
- Client - Server Model $\Rightarrow$ Application - Layer



Topic : DHCP Client – Server Scenario



- Typically DHCP server will be co-located in router
[serving all subnets to which routers is attached]
- DHCP runs over UDP ✓
[Client uses port 68 and server uses port 67]
- Sometimes a subnet may have multiple DHCP servers ✓

Host A
DHCP client (P₁)

Router
DHCP server (P₂)





Topic : DHCP Operation

DORA



→ DHCP protocol is four step process.

1. DHCP Server Discovery
2. DHCP Server Offer(s) ✓

3. DHCP Request

4. DHCP ACK





Topic : DHCP Server Discovery



- Newly arriving host (client) is to find DHCP server
- Client broadcast "DHCP discover" message

Application Layer PDU

MSG

Source IP Address	: [0.0.0.0]: 68
Destination IP Address	: [255.255.255.255]: 67
yourIP	: [0.0.0.0]
Type	: DHCPDISCOVER
transaction ID	: 752 ✓

FLAG=

UDP [DHCP MSG]

IP [UDP segment]

FRAME [IP Datagram]

- ① Source MAC Add = Host MAC Add.
- ② Dest. MAC Add = Broadcast MAC

DHCP Discover ⇒ (Broadcast)

[0.0.0.0]

: This Host

[255.255.255.255]

: Network / Sub-network Broadcast Address (limited Broadcast)



Topic : DHCP Server Offer



- When DHCP server receives "DHCP discover" message
- DHCP server responds by sending "DHCP offer" message

<u>Source IP Address</u>	:	[210 . 210 . 210 . 2]	:	67
<u>Destination IP Address</u>	:	[255 . 255 . 255 . 255]	:	68
<u>your IP</u>	⇒	[210 . 210 . 210 . 4]		
<u>Type</u>	:	DHCPOFFER		
<u>transaction ID</u>	:	[752]	✓	
<u>DHCP Server IP</u>	:	[210 . 210 . 210 . 2]		
<u>Lifetime</u>	:	3600	sec	

DHCP offer

Frame :-

Source MAC Add
= DHCP Server MAC

① Broadcast (default)
Dest. MAC Add.
= Broadcast MAC

② Unicast
Dest. MAC Add
= Host MAC Address



Topic : DHCP Request



- Newly arriving host (client) will choose from among one or more server offers
- DHCP Client responds to its selected offer with a "DHCP request" message

Source IP Address	: [0 . 0 . 0 . 0] : 68
Destination IP Address	: [255 . 255 . 255 . 255] : 67
yourIP	: [210 . 210 . 210 . 4]
Type	: DHCPREQUEST
transaction ID	: [753]
DHCP Server IP	: 210 . 210 . 210 . 2
Lifetime	: 3600 sec

DHCP Req.

- DHCP ACK (Positive)
- DHCP NAK (Negative)



Topic : DHCP Acknowledgment

→ DHCP server responds to DHCP client with a "DHCP ACK" message

<u>Source IP Address</u>	: [210 . 210 . 210 . 2] : 67
<u>Destinanton IP Address</u>	: [255 . 255 . 255 . 255] : 68
<u>yourIP</u>	: [210 . 210 . 210 . 4]
<u>Type</u>	: DHCPACK
<u>transaction ID</u>	: [753]
<u>DHCP Server IP</u>	: [210 . 210 . 210 . 2]
<u>Lifetime</u>	: 3600 sec



Topic : DHCP Operation



=> DHCP server can return more than just allocated IP address on subnet

→ Address of first hop router for client (Default Gateway)

→ Name and IP address of DNS server

→ Network Mask

=> DHCP server discovery and DHCP server offer(s) are optional steps
[if a client remember and wishes to reuse previously allocated network address]

Example :-

[MCQ]



#Q. DHCP messages always encapsulate into _____.

- ☐ A TCP Segment Payload Field
- ☒ B UDP Segment Payload Field
- ☐ C Directly in IP datagram Payload Field
- ☐ D None of the above

Ans: B

[MCQ] [GATE-IT-2006, 2-Marks]



#Q. A router has two full-duplex Ethernet interfaces each operating at 100 Mb/s. Ethernet frames are at least (84 bytes) long (including the Preamble and the Inter-Packet-Gap). The maximum packet processing time (in microseconds) at the router for wirespeed forwarding to be possible is _____.

A 0.01

✓ B 3.36

C 6.72

D 8

Min^m Frame length

= 84 byte (including overhead)



Time required to receive one min^m size frame

$$= \frac{\text{min}^m \text{Frame size}}{\text{Bandwidth}} = \frac{84 \text{ byte}}{100 \text{ Mb/sec}} = \frac{84 \times 8 \text{ bits}}{10^8 \text{ bits/sec}} = 6.72 \mu\text{s}$$

$$\text{Max}^m \text{Packet Processing time} = \frac{6.72 \mu\text{s}}{2} = 3.36 \mu\text{s}$$

Ans: B



Topic : Socket



→ One endpoint of a two-way communication

→ Two network process communicate via a pair of sockets

Network
Communication

Socket
Communication

→ Door or interface of a network process

→ Socket is data structure inside the program



Topic : Socket API



→ Set of library function calls
[Network (Socket) Communication]

→ Berkeley Sockets API

API : Application Programming Interface



Topic : Socket Programming

⇒ Network Programming PW

→ Client / server application that communicate using sockets

Process : Controlled by application developer ✓

OSI Model : Mostly controlled by operating system



Topic : Types of Sockets



Stream Socket : SOCK_STREAM

- Uses TCP protocol
- Connection-oriented service
- Reliable and byte stream oriented

Datagram Socket : SOCK_DGRAM

- Uses UDP protocol
- Connection-less service
- Unreliable datagram



Topic : Creating a Socket



```
#include <sys/types.h>
#include <sys/socket.h>
```

socket() → System Call
⇒ Req. to OS
for creation of
socket file

```
int socket ( int domain, int type, int protocol );
```

domain : Protocol Family
[PF_INET, AF_INET, PF_UNIX etc]

* type : Communication Protocol semantics
[SOCK_STREAM, SOCK_DGRAM]

⇒ Type of socket

protocol : Specify particular protocol, set "zero" as default

* return : On success, a file descriptor for the new socket
On error, -1 is returned

→ sock_fd

```
int a;
a = socket();
if(a == -1) Error;
```



2 mins Summary



Topic

DHCP ✓

Topic

Socket Programming →



THANK - YOU

